

Remnant Fieldworks Inc. — Master Summary

Science Research · Patents · Trademarks

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Proof Before Power™ — built to a world-class standard, for Christ.

0. How to read this document

This is the single reconciled record of everything Remnant Fieldworks (RF) has **proven** (science), everything RF has **filed to protect** (patents), and everything RF has **claimed as a mark** (trade-marks). It is written to the RF Standing Covenant: claims are kept **narrower than the evidence**, failures and gate-stops are **preserved honestly**, and every experimental result is traceable to a public GitHub commit and a Zenodo DOI.

Three reinforcing pillars govern the whole program:

The experiments build evidence and credibility. The patents protect invention. The pilots prove customer value.

All three reinforce one another without overstating what any one establishes.

PART I — SCIENCE RESEARCH

RF's public scientific corpus spans **five research programs**, all preregistered, executed, and published with raw-data provenance (GitHub + Zenodo DOI, CC BY 4.0).

Program	Theme	Substrate	Status
ARK	ExecutionProof authorization boundary (Verification Before Execution)	Quantum hardware (441-448) + classical software (449-492)	38 experiment IDs, 60 records — active
WITNESS	Quantum-sourced authorization nonces with verifiable provenance	IBM quantum hardware (<code>ibm_fez</code>)	2 experiments, 6 cases — complete + published
BELLWETHER	Nonclassicality (Bell/Mermin/contextuality) witness bound into a nonce	IBM quantum hardware (<code>ibm_fez</code>)	3 experiments — complete + published
CHRONO	Temporal nonclassicality (Leggett-Garg) witness bound into a nonce	IBM quantum hardware (<code>ibm_fez</code>)	1 experiment — complete + published
UIP Phase 1	Foundational physics program (precursor)	—	Closed

Corpus totals (ARK + WITNESS, reconciled 2026-07-18): 60 case records across 38 experiment IDs → **57 PASS · 2 honest FAIL · 1 GATE-STOP, 56/56 Zenodo publications complete (100%)**. P02 Latency/Throughput/Scale (ARK-483-492) complete 10/10. The BELLWETHER (3) and CHRONO (1) quantum-witness experiments are additionally published (4 more DOIs).

1. ARK Series — ExecutionProof Authorization Boundary

Core claim tested (VBE): Permission at approval time is not permission at execution time. Authorization must be re-verified against current state at the **moment of execution**.

1.1 Hardware Quantum ARKs (ARK-441-448) — 9 records

Backends: ARK-441 `ibm_kingston` ; ARK-442-448 `ibm_marrakesh` (all 156-qubit Heron r2).

ARK	Question	Verdict	DOI
441	VBE authorization boundary baseline	✓ PASS	10.5281/zenodo.21404867
442	Delay / expiry boundary (stale authority fails closed)	✓ PASS	10.5281/zenodo.21404867
443	Two-of-three quorum (single compromised authorizer)	✓ PASS	10.5281/zenodo.21404867
444	Decision-to-execution integrity (tampered action caught)	✓ PASS	10.5281/zenodo.21404867
445	Tri-state ALLOW/HOLD/DENY	✗ FAIL (honest)	10.5281/zenodo.21404867
445b	Reset-free re-test	✓ PASS	10.5281/zenodo.21418404
446	Cross-device replication	✓ PASS	10.5281/zenodo.21404867
447	Pauli twirling vs baseline	✓ PASS	10.5281/zenodo.21404867
448	Dynamical decoupling vs baseline	⊖ GATE-STOP	10.5281/zenodo.21404867

Honest-record note: ARK-446 has no MANIFEST.txt in the repo (only RESULTS.pdf + PREREGISTRATION.pdf were deposited). Preserved as-executed.

1.2 Classical / Software ARKs, pre-P01 (ARK-449-457) — 10 records

Substrate: pure classical software.

ARK	Question	Verdict	DOI
449	State change after verification	✓ PASS	10.5281/zenodo.21406335
450	Verified-action substitution attack	✓ PASS	10.5281/zenodo.21420310
451	Authority revocation during execution	✓ PASS	10.5281/zenodo.21419480
452	Multi-step workflow, one invalid step	✓ PASS	10.5281/zenodo.21406668
453	Conflicting evidence must HOLD	✓ PASS	10.5281/zenodo.21419198
454	Self-approval & circular delegation	✓ PASS	10.5281/zenodo.21418935
455	ProofRecord tamper (dual-verifier)	✗ FAIL (honest)	10.5281/zenodo.21418388
455b	ProofRecord tamper — corrected retest	✓ PASS	10.5281/zenodo.21418404
456	Fail-closed dependency loss	✓ PASS	10.5281/zenodo.21420107
457	Cross-context authorization replay	✓ PASS	10.5281/zenodo.21421742

1.3 P01 Production-Boundary Series (ARK-458-482) — ✓ COMPLETE

25/25 experiments PASS. Real, end-to-end authorization boundaries across 5 action domains × 5 failure modes. Dual independent guards (V1 JavaScript + V2 Python).

#	Series (action tuple)	ARKs	Verdicts	DOIs (version)
1	Cloud IAM Role Grant (principal, role, re-source_arn, account, grant_action)	458-462	5 PASS	21432645, 21432879, 21432883, 21432887, 21432889
2	Production Deployment (service, environment, version, deploy_type, approver)	463-467	5 PASS	21433070, 21433072, 21433074, 21433077, 21433081
3	Database Destructive Query (database, table, operation, schema_ver, exec_mode)	468-472	5 PASS	21433449, 21433451, 21433453, 21433455, 21433457
4	Financial Transaction (from, to, amount, currency, txn_type)	473-477	5 PASS	21433459, 21433461, 21433463, 21433465, 21433469
5	API Rate Limit (api_key, endpoint, rate_limit, window, tier)	478-482	5 PASS	21433473, 21433476, 21433479, 21433483, 21433485

Each series tests the same 5 failure modes: **Exact-Action Binding · Revocation At Execution · Dependency Loss · Cross-Context Replay · Human Escalation.**

P01 aggregate metrics:

- Total decisions tested: **20,000**
- Verdicts: **25 PASS (100%)**
- Dual-guard concordance: **20,000 / 20,000 (100%)**
- Kill-gate falsifiability: **5,306 / 5,306 wrong-allows detected (100%)**
- All gate metrics met: $g_{\text{allow}} = 1.0000$, $g_{\text{deny/hold_min}} = 1.0000$, $g_{\text{margin}} \geq 0.9500$

1.4 P02 Latency / Throughput / Scale Series (ARK-483-492) —

10/10 COMPLETE

CUT = frozen ARK-458 guard (Verification Decision) + minimal in-memory reference Authority/
Evidence engines (measurement only, **not** production engines).

ARK	Question	Verdict	Result	DOI
483	Verification decision — latency	✓ PASS	worst-path p95 1.822 µs (Py) / 0.652 µs (JS); ceiling 1000 µs	10.5281/zenodo.21432647
484	Verification decision — burst throughput	✓ PASS	1.66M dec/s (Py, 16.6× pred) / 4.52M dec/s (JS, 30.2× pred)	10.5281/zenodo.21433111
485	Verification decision — sustained throughput	✓ PASS	1.50M dec/s (Py) / 9.52M dec/s (JS), 100% acc over 60s	10.5281/zenodo.21434398
486	Verification decision — cost at scale	✓ PASS	fixed prior FAIL ; Scenario B realistic \$7.47e-06/M (Py) / \$1.18e-06/M (JS); naive \$0.20/M disclosed	10.5281/zenodo.21434400
487	Authority Engine — cold start	✓ PASS	p95 9.34 ms (mean 7.87 ms); correctness gate PASS	10.5281/zenodo.21434402
488	Authority Engine — p95 latency	✓ PASS	warm p95 0.32 µs , p99 0.41 µs (200k decisions)	10.5281/zenodo.21434405
489	Authority Engine — burst throughput	✓ PASS	3.09M dec/s , 100% acc	10.5281/zenodo.21434407
490	Authority Engine — sustained throughput	✓ PASS	2.49M dec/s over 60s, 100% acc (~81% of burst)	10.5281/zenodo.21434409
491	Authority Engine — cost at scale	✓ PASS	Scenario B \$3.59e-06/M ;	10.5281/zenodo.21434411

ARK	Question	Verdict	Result	DOI
			naive \$0.20/M disclosed	
492	Evidence Engine — cold start	✓ PASS	p95 44.0 ms ; tamper + broken-chain correctly DENY	10.5281/zenodo.21434413

Honest findings: (a) DENY slower than ALLOW in ARK-483 (reason-string formatting dominates tail). (b) ARK-486 corrected a cost-model category error (per-request serverless price per in-process decision) → verdict on realistic running-service model, naive bound disclosed. (c) ARK-487–492 measure minimal in-memory reference engines, not production engines; claims bounded to single-threaded in-memory load. All disclosed.

2. WITNESS Series — Quantum-Sourced Authorization Nonces

Repo: github.com/derekhone/witness-testbeds · Concept DOI 10.5281/zenodo.21424323

Exp	Title	Hardware	Result	DOI (version)
WITNESS-1	Quantum-sourced nonces with verifiable provenance	ibm_fez (Heron r2), job d9dg-p6kjeosc73f-higsg , 4000 shots, 256 unique outcomes	3/3 PASS (honest verify · tamper detect · replay prevention)	10.5281/zenodo.21424324
WITNESS-2	Length-prefixed nonce with record-hash field integrity	ibm_fez , job d9di7nkin-v1c73ap4ed0 , 4000 shots, 256 unique outcomes	3/3 PASS (9-check honest verify · 4/4 tamper · cross-context replay)	10.5281/zenodo.21425381

WITNESS-2 `quantum_nonce` = e425dc92...659b ; `record_hash` = 271ff5ea...7ae1 .

3. BELLWETHER Series — Nonclassicality Witness Bound Into a Nonce

Repo: github.com/derekhone/bellwether-testbeds . Physical nonclassicality (spatial/measurement) cryptographically bound into an authorization nonce/ProofRecord as a tamper-evident entropy witness.

Exp	Witness	Bound	Measured	Hardware	DOI (version)
BELLWETH-ER-1	Bipartite CHSH	classical ≤ 2 ; Tsirelson 2.828	S = 2.514 ($>3\sigma$)	ibm_fez , job d9d-je2cjeosc73f hm230 , 4000 shots	10.5281/zenodo.21430442
BELLWETH-ER-2	Multipartite Mermin-3 (GHZ)	classical ≤ 2 ; quantum max 4	 M = 3.423 ($n_\sigma \approx 61.6$)	ibm_fez , job d9dno91ht-sac739da4d0 , 8000 shots	10.5281/zenodo.21430446
BELLWETH-ER-3	State-independent contextuality (Peres–Mermin)	noncontextual ≤ 4 ; quantum 6	$\chi = 5.268-5.376$ across 3 states (all >4 , all 3σ)	ibm_fez , job d9do1s1ht-sac739dahpg , 18000 shots	10.5281/zenodo.21430451

Authorization-model arc: BW1 bipartite → BW2 multipartite (unforgeable by any pair) → BW3 state-robust (holds regardless of preparation). **BW3 is the broadest / most patent-worthy** of the family (state-robust authorization).

4. CHRONO Series — Temporal Nonclassicality Witness

Repo: github.com/derekhone/chrono-testbeds . Three-time **Leggett-Garg** inequality K3 on a single qubit, bound into an authorization ProofRecord — maps directly to “Verification Before Execution” in the time dimension.

Exp	Bound	Measured	Hardware	DOI (version)
CHRONO-1	macrorealist K3 ≤ 1 ; quantum (Lüders) max 1.5	K3 = 1.450 ($>3\sigma$, 97% of ideal)	ibm_fez , job d9dkj6ineu4c739 nddrg , 6000 shots	10.5281/zenodo.21430455

Together the quantum-witness programs span **all three nonclassicality families**: nonlocality (BW1/BW2), contextuality (BW3), and temporality (CHRONO).

5. Scientific integrity boundary (carried into every artifact)

- Results apply to the **specific backends, qubit pairs, calibration snapshots, shot counts, and parameters** used.

- These are hardware noise-characterization studies and software-harness tests — **not cryptographic security proofs**.
- Error **mitigation** is not error **correction** (no QEC).
- Quantum witnesses (BELLWETHER/CHRONO) are **device-dependent**; locality, detection, compatibility, and clumsiness loopholes remain **open**. Not loophole-free; not device-independent certified randomness.
- Honest FAILs (ARK-445, ARK-455) and the ARK-448 GATE-STOP **stand** alongside clean retests — that preserved record is itself the credibility asset.

PART II — PATENTS

Portfolio scale (as of 2026-07-18): 56 USPTO filings. Derek Adam Hone is **sole inventor**.

Layer	Count	Detail
Nonprovisional utility	8	Parent 19/529,283 (filed Feb 4, 2026) + 7 continuations-in-part (filed Feb-Jul 2026)
Provisional	48	47 provisionals filed Jan 10–12, 2026 + Commercial Gate 63/971,820 (Jan 30, 2026)
Total	56	Priority for the classical governance / crypto / authorization stack locked before any quantum publication

2.1 Named patent-family connections (evidence ↔ invention)

- **No-self-approval patent** — supported by the ARK-454 experimental design (self-approval, delegated, and circular delegation protection).
- **ProofRecord™ integrity** — ARK-455 / ARK-455b directly tested tamper-resistance of the machine-readable, independently verifiable decision record.
- **Three-state control model (ALLOW / HOLD / DENY)** — HOLD as a first-class conservative outcome, evidenced by ARK-445 / ARK-445b (HOLD metrologically separable from ALLOW and DENY).
- **Fail-closed dependency handling, exact-action binding, context binding** — evidenced across ARK-456, ARK-457, and the full P01 production series.

2.2 How experiments support the patent program (and the hard limit)

Experiments create **dated evidence that disclosed systems actually work** — mapping real implementations to existing claims, surfacing narrower dependent-claim opportunities, and distinguishing **working examples** from **prophetic examples**.

HARD LIMIT: An experiment performed after filing **cannot repair missing disclosure or add new matter** to an existing application. New matter needs a **continuation-in-part or a new application** with its own later priority date. Actual results are always labeled **working examples** — never implied to have existed at an earlier filing date.

2.3 Quantum IP — 12-month defensive-publication posture

The classical stack is protected by the filed provisionals/nonprovisionals above (priority locked before any quantum disclosure). Because budget cannot afford additional filings right now, the quantum layer (**WITNESS, BELLWETHER, CHRONO**) switched to **defensive publication**: a timestamped Zenodo DOI = prior art protecting freedom to operate, at zero cost.

Doctrine: “File-first when we can; publish-to-protect when we can’t.”

- All quantum publications dated **2026-07-18** → **US grace-period deadline 2027-07-18** (practical cutoff ~2027-06-18). Ex-US rights are likely already barred by publication.
- If exactly one filing is made: a single **small-entity US provisional on the genus** (any-nonclassicality-witness-bound authorization nonce), with **BELLWETHER-3 (contextuality) as lead example** — the broadest, state-robust claim.
- This cannot be bolted onto 19/529,283 (would be new matter) — it needs its own provisional/CIP.
- Full tracking: `RF-Quantum-IP-File-Within-12-Months-Ledger.md` (+ PDF/DOCX).

2.4 The IP Gate (run before every public disclosure)

Before anything goes to GitHub / Zenodo / LinkedIn / RF-100, each result must pass:

1. Does it merely demonstrate something **already disclosed** in a filed application? → safe to publish (working-example evidence).
2. Does it reveal an **improvement / new configuration / unexpected result**? → potential new matter.
3. Is that improvement **already supported by a pending application**? → if yes, publish; if no, treat as unprotected.
4. Should we **file before publishing**? → if new matter, **YES**.
5. Can we **publish the result while withholding unnecessary implementation details**?

PART III — TRADEMARKS

Corrected, authoritative status (Derek, 2026-07-18): five marks are the subject of **filed, pending federal trademark applications** with the USPTO. Use the ™ symbol; do **not** imply registration (®) until the USPTO grants.

Mark	Role	Status
Remnant Fieldworks™	Company	Pending federal TM application
ExecutionProof™	Commercial platform (VBE doctrine)	Pending federal TM application
Proof Before Power™	Doctrine / brand line	Pending federal TM application
Verification Before Execution™	Framework	Pending federal TM application
ProofRecord™	Branded, machine-readable decision artifact	Pending federal TM application
VaultProof™	Implementation layer	Common-law mark only — NOT a filed federal application

Language discipline: describe the first five as “pending federal trademark applications” with TM. For **VaultProof™**, only ever claim **common-law** use — never write “filed” or “registered ®” until a separate filing is confirmed.

PART IV — COMPANY ARCHITECTURE & INFRASTRUCTURE

Stable architecture

- **Remnant Fieldworks™** — company (pending TM)
- **Proof Before Power™** — doctrine (pending TM)
- **Verification Before Execution™** — framework (pending TM)
- **ExecutionProof™** — commercial platform, proprietary (pending TM)
- **ProofRecord™** — branded artifact, proprietary (pending TM)
- **VaultProof™** — implementation layer (common-law)
- **RF-100** — implementation-neutral, publicly reviewable standard: <https://conceptradar.notion.site/RF-100-4cf29f7e8ad84cddb7dea7c4126e94c>

Public repositories

Repo	Program	Status
executionproof-testbeds	ARK corpus (canonical)	Live, main
witness-testbeds	WITNESS 1-2	Complete + published
bellwether-testbeds	BELLWETHER 1-3	Complete + published
chrono-testbeds	CHRONO-1	Complete + published
vaultproof-agent-guard	VaultProof implementation	Live (PR #4 open)
uip-phase1-testbeds	UIP Phase 1	Closed

Concept DOIs

- ARK concept: 10.5281/zenodo.21398675
- WITNESS concept: 10.5281/zenodo.21424323
- BELLWETHER concept: 10.5281/zenodo.21430441 (BW1) / 21430445 (BW2) / 21430450 (BW3)
- CHRONO concept: 10.5281/zenodo.21430454

Provenance model

Commits are timestamped but **not** GPG/SSH-signed. Provenance = public commit history + MANIFEST SHA-256 preregistration locks + Zenodo DOIs. VaultProof app-layer ed25519 ProofRecord signatures are distinct from git commit signing.

PART V — CONSOLIDATED SCOREBOARD

Program	Experiments	Records	PASS	FAIL	GATE-STOP	Published
ARK Hardware (441-448)	9	9	7	1	1	9/9 ✓
ARK Classical pre-P01 (449-457)	10	10	9	1	0	10/10 ✓
ARK P01 Production (458-482)	25	25	25	0	0	25/25 ✓
ARK P02 Latency/Throughput/Scale (483-492)	10	10	10	0	0	10/10 ✓
WITNESS (1-2)	2	6	6	0	0	2/2 ✓
ARK+WITNESS sub-total	56	60	57	2	1	56/56 ✓
BELLWETHER (1-3)	3	3	3	0	0	3/3 ✓
CHRONO (1)	1	1	1	0	0	1/1 ✓
GRAND TOTAL	60	64	61	2	1	60/60 ✓

IP totals: 56 USPTO patent filings · 5 pending federal trademarks + 1 common-law mark.

PART VI — FORWARD ROADMAP (abbreviated)

The **500-experiment schema (ARK-458-957)** is a floor, not a ceiling — 10 priority groups (P01-P10), with new groups anticipated beyond P10.

- **P01 Production Boundary (458-482)** —  COMPLETE
- **P02 Latency/Throughput/Scale (483-492)** —  complete (10/10 PASS, all published); next P03 Dependency Cascade (ARK-493+)
- **P03 Reliability/Failure Modes (508-532)** — next
- **P04-P10** — ProofRecord & crypto, identity/delegation, evidence policy, AI-agent tool-use, treasury/financial, VaultProof/digital-asset, cloud control-plane

Operating rhythm: ~1 meaningful, honestly-published experiment per week + 10-20 targeted commercial conversations. **70/30 rule:** 70% pilot acquisition & delivery, 30% experimentation supporting customers, product, standards, or IP.

Stewardship signal

Strength lies in the preservation of failures, honored gates, corrected language, the clean separation of open standard from proprietary product, and a fully challengeable public record. The next faithful move is patience, security, truthfulness, company protection, and one narrow service act proving the boundary protects a real organization.

 Soli Deo Gloria — world-class work, offered for Christ.

This summary reconciles the RF Master Corpus Scoreboard, Infrastructure Record, Strategic Doctrine, Synthetic-Phase Closure, and the BELLWETHER/CHRONO quantum-witness records as of 2026-07-18. Claims are held narrower than the evidence. All experimental outcomes — PASS, FAIL, HOLD, and GATE-STOP — are preserved honestly.